

Yingjie Lei

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Education

Johns Hopkins University <i>M.S.E. in Computer Science</i>	Aug. 2026 – Present
South China Normal University – Aberdeen Institute of Data Science & AI <i>B.Eng. & B.Sc. (Hons) in Artificial Intelligence</i>	Sep 2022 – Jun 2026
◦ Dual-degree program in Artificial Intelligence ◦ CN Degree: B.Eng. in Artificial Intelligence, South China Normal University (Project 211) ◦ UK Degree: B.Sc. (Hons) in Artificial Intelligence, University of Aberdeen; graduated with First Class Honours ◦ GPA: 90.9/100 (CN), 20.23/22 (UK); Rank: 5/88 (Top 5.7%) ◦ Core Courses: Machine Learning, Data Mining and Visualization, Robot Technology, Languages and Computability, Discrete Mathematics, Probability and Mathematical Statistics	

Research Interests

World models, representation learning, and embodied AI.

Publications & Preprints

PrefBench: Evaluating Zero-Shot LLM Agents in Hidden-Preference Personalized Pricing Negotiations ↗	May 2026
Yingjie Lei <i>arXiv preprint, arXiv:2605.22855, 2026.</i>	
Draft-and-Target Sampling for Video Generation Policy ↗	Under Review
Qikang Zhang, Yingjie Lei , Wei Liu, Daochang Liu Under review, 2026.	
Test Time Training for 4D Medical Image Interpolation ↗	Apr. 2025
Qikang Zhang*, Yingjie Lei *, Zihao Zheng, Ziyang Chen, Zhonghao Xie IJCNN 2025, Oral Presentation. *Equal contribution.	

Research Experience

Representation-to-Decision Reliability in World Models <i>Independent Research</i>	Jun. 2026 – Aug. 2026
◦ Developed the Planning Metric Adapter (PMA) for frozen DINO-WM and Control-Aligned Latent Models (CALM) for jointly training LeWM's representation, action-conditioned dynamics, and control projection. ◦ Evaluated three CALM costs on four tasks using three seeds and the same 50 queries per seed, revealing task- and interface-dependent closed-loop behavior. ◦ Found that prediction error, candidate ranking, and closed-loop planning success can move in different directions; predictive fidelity and decision utility are related but distinct. ◦ On 48 PushT queries, found useful ordering in realized endpoints but near-chance ordering after imagined rollout, localizing one failure mode to the transition stage. ◦ Synthesized the system design, experiments, and failure analysis into a research manuscript.	
Undergraduate Thesis: Personalized Pricing of Customized Car Features <i>Supervised by Dr. Binod Bhattacharai; Thesis grade: 92/100; Outstanding Thesis (Top ~3%)</i>	SCNU / Aberdeen Jan. 2026 – Apr. 2026
◦ Formulated hidden-preference personalized pricing as a POMDP with latent buyer traits, multi-round negotiation, and profit-oriented sequential decision making.	

- Built a reproducible simulator benchmark with public-data-grounded persona generation, explicit observable/hidden trait separation, fixed evaluation splits, and shared episode-level metrics.
- Implemented heuristic and learning-based seller agents, including **PPO** and **DreamerV3**, and ran ablations on reward design, semantic features, and inference-time adaptation.
- Refined the thesis into [PrefBench](#), a preprint for evaluating LLM agents under hidden buyer preferences.

Draft-and-Target Sampling for Video Generation Policy

SCNU

Second Author, manuscript under review

Apr. 2025 – Sep. 2025

- Proposed **Draft-and-Target Sampling (DTS)**, a training-free acceleration framework for diffusion-based video generation that uses dual denoising trajectories within a single model.
- Introduced **Token Chunking** and **Progressive Acceptance** to reduce redundant resampling while preserving output fidelity.
- Achieved up to **2.1× speed-up** on *Meta-World*, *iThor*, *Libero* with minimal accuracy drop, and surpassed baselines in accuracy on *iThor*.
- Studied the method in embodied-style video-policy settings, emphasizing inference efficiency for downstream decision-making rather than only generic video generation quality.
- **Personal Contributions:** Implemented and refined the core pipeline, ran ablations and benchmarks on *Meta-World* and *iThor*, organized results, created task visualizations, and contributed to manuscript writing and revision.

Test Time Training for 4D Medical Image Interpolation

SCNU

Co-first Author, paper published in the Proceedings of IJCNN 2025 (Oral)

Sep. 2024 – Feb. 2025

- Proposed a **Test Time Training (TTT)** framework for 4D medical image interpolation to handle distribution shift without requiring target-domain labels.
- Designed a shared-encoder adaptation mechanism using self-supervised tasks, including *rotation prediction* and *3D-MAE reconstruction*, during inference.
- Achieved state-of-the-art results on **Cardiac MRI** and **4D-Lung CT** (**33.7dB** and **34.0dB PSNR**) with improved robustness under distribution shift.
- Framed the method as a reusable adaptation template for downstream medical vision tasks such as segmentation and registration under shifting test distributions.
- **Personal Contributions:** Implemented the core 3D-MAE module, ran experiments and metric evaluation, created visualizations, improved code clarity, and contributed to manuscript writing and revision.

Honors & Awards

SCNU Outstanding Undergraduate Thesis (Top ~3%)	Jun 2026
Selected among the top undergraduate theses in the graduating cohort	
SCNU Comprehensive Scholarship (3rd Prize)	Apr 2026
Merit-based award recognizing strong academic record and overall performance	
SCNU Outstanding Student Scholarship	Mar 2025
Awarded to top students with excellent academic performance (Top 8% in cohort)	
SCNU-UoA Joint Institute Scholarship (3rd Prize)	Oct 2024
Granted for outstanding GPA performance, ranked 5th out of 86 students	

Skills & Qualifications

- **Programming & ML:** Python, PyTorch, Hugging Face, Transformers, diffusion models, reinforcement learning, scikit-learn.
- **Research Tools:** Git, LaTeX, Jupyter, NumPy, pandas, Matplotlib, Seaborn, Zotero.
- **Tests & Languages:** IELTS Academic 7.5 (L 8.5, R 8.5, W 6.5, S 7.0); GRE 319 (Q 170, V 149, AW 3.5); Mandarin and Cantonese (Native), English (Fluent).